

Võ Hoàng Ngân

Software Engineer

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OBJECTIVE

A high-achieving Computer Science student at HCMUT with a solid 3.8/4.0 GPA and a strong logical foundation (National Math Olympiad winner). Proficient in Full-stack development, with hands-on experience in optimizing data pipelines and researching autonomous navigation algorithms. Seeking a Software Engineer Fresher to apply analytical problem-solving skills to real-world technical challenges, aiming to evolve into a Senior Software Engineer within the next 5 years.

EDUCATION

Ho Chi Minh City University of Technology – Vietnam National University, Ho Chi Minh City (HCMUT – VNU-HCM) 2023 - Present

Major: Computer Science

GPA: 3.8/4

Le Hong Phong High School for the Gifted, Ho Chi Minh City 2020 - 2023

Math-specialized class

TECHNICAL SKILLS | EXPERIENCE

Mathematics Tutor	Mentored and coached students to achieve City-level awards (Second Prizes) in Mathematics competitions.
Research Assistant Institute of Mathematics and Computing Science (IMACS)	Conducted in-depth research on algorithm of non-GPS drones.
Languages	C++ (Proficient), Python, Java, JavaScript (ES6+), HTML5, CSS3.
Frameworks/Libraries	ReactJS, Node.js, Express.js.
Databases	MySQL, PostGreSQL
Tools & Others	Git & GitHub, Figma.
CS Fundamentals:	Operating Systems, Data Structures & Algorithms, Computer Networks.
Soft Skills	Strong self-learning capability (proven by rapid GPA improvement), Team Management, and effective communication in English.

PROJECTS

Tutor Support System | https://github.com/Kras2508/Tutor_Support_System.git | Team size: 6 9/2025 - 12/2025

Fullstack Developer

- Developed RESTful APIs for the Tutor module, enabling tutors to manage availability slots, track student progress, and view session history.
- Implemented Complex Business Logic: Built robust algorithms for time slot overlap validation and handled booking cancellation workflows effectively.

- **Database Optimization:** Leveraged MongoDB Aggregation Pipeline to perform complex data joins and generate real-time statistics for the "My Students" dashboard.
- **Data Integrity & Validation:** Integrated **Joi** middleware for strict input validation and implemented defensive coding practices to handle missing or inconsistent data fields.
- **Integration:** Successfully integrated and deployed the database on MongoDB Atlas, ensuring secure environment configuratio
- **Technologies Used:** Node.js, Express.js, MongoDB (Mongoose), RESTful API.

Bright Shoes | bright-shoes.vercel.app | Team size: 505/2025 - 08/2025

Database Designer

- **Database Schema Design:** Designed database schema using ERD and applied 3NF normalization to ensure data integrity.
- **Technologies Used:** ReactJS, Vite, Node.js, Express.js, MySQL, VNPay API, Vercel.

Doctor Landing Page | <https://ngankaka.github.io/Doctor/> | Team size: 16/2024 - 7/2024

Frontend Developer

- Developed an interactive user interface from a Figma design, ensuring a pixel-perfect match.
- Implemented interactive UI elements and smooth scrolling animations using modern JavaScript (ES6+).
- **Technologies Used:** HTML5, CSS3, JavaScript.

EXPERIENCE

AIPOWER | Full-stack Developer Intern Ho Chi Minh City, Vietnam | Jan 2026 – April 2026

Project: Lopas | Role: Backend Engineer

- Developed an automated 3D point cloud processing pipeline for object reconstruction.
- Engineered auto-alignment and symmetry-based algorithms to reconstruct regions and correct object skewness.
- Tech Stack: FastAPI, Vue 3, TypeScript, Three.js, Vite, Docker.

Institute of Mathematics and Computing Science (IMACS) | Research Assistant | Jan 2026 – Present

- Autonomous Navigation: Researched and developed localization algorithms for non-GPS drones in complex, GPS-denied environments using C++.
- Algorithmic Optimization: Designed mathematical models for Inertial Navigation Systems (INS) and Sensor Fusion to mitigate cumulative positioning errors and enhance navigation precision.

Mathematics Tutor

- Mentored students to win Second Prizes in City-level Mathematics competitions.

HONORS & AWARDS

Academic: Academic Encouragement Scholarship - 120% Scholarship (Ranked 2nd in Faculty) , OISP Scholarship , Odon Vallet Scholarship.2024, 2025

National: Third Prize (2022) & Honorable Mention (2021) in the Vietnam Mathematical Olympiad (VMO).2022, 2023

International: Silver Medal - Asia International Mathematical Olympiad (AIMO) , Gold Medal - Hong Kong International Mathematical Olympiad (HKIMO), Aluminium Compass Prize in Iranian Geometry Olympiad (IGO).2022, 2023

City-level: First Prize in HCMC Mathematical Olympiad (2 consecutive years), First Prize in Ho Chi Minh City Handheld Calculator Mathematics Competition.2022,2023

CERTIFICATIONS & INTERESTS

Certifications: IELTS Academic 7.0 , MOS Excel 2019.2024

Interests: Competitive Programming (LeetCode), Strategy Board Games (Chess), and Football, Badminton.